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Interrupted time-series analysis yielded an effect estimate concordant with the cluster-randomized controlled trial result

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Abstract

Objective: We reanalyzed the data from a cluster-randomized controlled trial (C-RCT) of a quality improvement intervention for prescribing antihypertensive medication. Our objective was to estimate the effectiveness of the intervention using both interrupted time-series (ITS) and RCT methods, and to compare the findings.

Study Design and Setting: We first conducted an ITS analysis using data only from the intervention arm of the trial because our main objective was to compare the findings from an ITS analysis with the findings from the C-RCT. We used segmented regression methods to estimate changes in level or slope coincident with the intervention, controlling for baseline trend. We analyzed the C-RCT data using generalized estimating equations. Last, we estimated the intervention effect by including data from both study groups and by conducting a controlled ITS analysis of the difference between the slope and level changes in the intervention and control groups.

Results: The estimates of absolute change resulting from the intervention were ITS analysis, 11.5% (95% confidence interval [CI]: 9.5, 13.5); C-RCT, 9.0% (95% CI: 4.9, 13.1); and the controlled ITS analysis, 14.0% (95% CI: 8.6, 19.4).

Conclusion: ITS analysis can provide an effect estimate that is concordant with the results of a cluster-randomized trial. A broader range of comparisons from other RCTs would help to determine whether these are generalizable results.

Keywords: Interrupted time-series analysis; Randomized controlled trial; Health systems; Evaluation; Research methods; Comparative effectiveness

1. Introduction

Rigor and feasibility are often conflicting goals in the choice of study designs for evaluating health system interventions [1]. Randomized trials may be difficult to conduct for both practical and political reasons [2,3]. Simple but weak evaluation designs, such as comparing outcomes before and after an intervention, are often feasible but may yield misleading findings [4,5]. The high cost and slow pace of randomized, controlled trials (RCTs) can also impede their use in health systems research [6], although they sometimes may be quicker and less costly than observational studies [7]. Unfortunately, little is known about the relative validity of different approaches to evaluating the effectiveness of health system interventions, beyond theoretical arguments. The need for more empirical evidence on the validity of findings from randomized and nonrandomized study designs was recently emphasized by the Methodology Committee of the Patient-Centered Outcomes Research Institute [8].

In one such design, interrupted time-series (ITS) analysis, data collected at multiple time points before and after an intervention are evaluated to determine whether the intervention produced a discontinuity (change in level or slope) in comparison with the underlying secular trend [9]. This method is promoted as "a particularly strong quasi-experimental alternative to randomized designs when the latter are not feasible" [10, p. 172]. Frequently, the data used in ITS analyses of health system interventions are collected routinely as part of the delivery of care or the reimbursement process. Thus, this approach may be an attractive option that is both feasible and rigorous. However, to

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What is new?

- An interrupted time-series analysis yielded an effect estimate that is concordant with a conventional cluster, randomized, controlled trial analysis.
- This finding provides empirical support for the hypothesis that interrupted time-series analyses can provide reliable estimates of the effects of health system interventions.
- A broader range of comparisons would help to determine whether the findings from this study are generalizable.

our knowledge, no empirical studies have explored whether effect estimates from an RCT would be replicated by ITS analyses of the same data.

Between April 2002 and December 2003, we (A.F. and A.O.) conducted a cluster-randomized, controlled trial (C-RCT) with primary care physicians in Norway:—the Rational Prescribing in Primary Care (RaPP)—trial [11]. The trial was financed by the Norwegian Ministry of Health, motivated by the increased use of newer, expensive drugs for cardiovascular risk reduction, such as calcium channel blockers and angiotensin receptor blockers that are generally no more effective than older, less expensive medications.

We developed a multifaceted intervention, including educational outreach visits and computerized reminders, to encourage increased use of recommended antihypertensive medication (low-dose diuretics) [11]. A total of 146 primary care practices from two areas of Norway were recruited (participation rate, 38%), half of which were randomized to receive the intervention. We block randomized within two geographical areas to ensure balance in the number of practices in the intervention and control groups. The size of the blocks varied randomly between two, four, and six. The allocation list was generated by a colleague not directly involved in the research project with software from http://www.randomization.com. We gave our colleague identification numbers that represented each recruited practice, and she informed us whether the practice was allocated to the intervention or control group [11].

Practices in the intervention group were visited by a trained pharmacist who presented the physicians with recently published clinical guidelines that included a recommendation to use low-dose diuretics as first-line antihypertensive medication. We installed a software package during the visit that enabled extraction of data from the electronic medical record system and immediate feedback to the physicians regarding their current prescribing practices. The software also provided on-screen reminders about recommended first-line medication, which were triggered when patients returned after any recording of high blood pressure. Outcome data were collected from electronic medical records. The flow of participating practices is shown in Fig. 1.

The intervention proved successful. The trial showed that prescribing the recommended antihypertensive drugs doubled in the intervention group compared with the control practices [11]. The analysis followed the conventional approach of comparing outcomes in the two groups aggregated over the intervention period, while adjusting for differences between the groups during the preintervention period.

After the trial was completed, we questioned whether the results would have been similar if we had used ITS analysis instead of conducting an RCT. The prescription data we collected for this trial were dated, providing an opportunity to investigate this question by analyzing the data with both methods. The objective of study was to compare the effect estimates of these two approaches.

2. Methods

We reanalyzed the primary outcome from the trial with data most suitable for ITS analysis: prescribing low-dose diuretics. The data from the trial included all prescriptions for patients initiated on antihypertensive medication during the year before and the year after the launch of the intervention. To accommodate the ITS analysis, we decided to exclude a small number of prescriptions (n = 178) from more than 12 months before or more than 12 months after the intervention, which had been included in the original trial analysis. We did this before running the analyses.

The C-RCT effect estimate was recalculated by comparing the postintervention proportions of recommended prescribing in the intervention and control groups, using baseline prescribing levels for each practice as covariates (analysis of covariance) in a generalized estimating equation model incorporating all observations (prescriptions) while controlling for clustering effects—a recommended approach to analysis of cluster trials [12]. The reanalysis of the C-RCT was done to ensure that the same data were used for the C-RCT and the ITS analysis, and that the same metric was used for effect estimates (absolute risk differences).

Initially, we conducted the ITS analysis using data only from the intervention arm of the trial, because our main objective was to compare the findings from a single-group ITS analysis with the findings from the C-RCT. This addressed our original research question directly: What would the findings have been had we conducted a simple, uncontrolled ITS analysis rather than a C-RCT? All prescriptions were organized according to the number of days before or after the intervention took place, from which we constructed a time series of monthly rates of prescribing of

Fig. 1. Flow of practices in the RaPP-trial [11].

low-dose diuretics aggregated across all practices (i.e., a total of 24 data points). The number of prescriptions per month was more than 100, which is recommended as a minimum number of observations per time point for ITS analysis [13].

We used segmented regression methods to estimate changes in level or slope coincident with the intervention, controlling for baseline trend (during 12 preintervention months) [13]. The intervention effect was assumed to occur immediately after implementation, so we decided not to include a transition period in the ITS analysis. To combine both the immediate impact (level change) and the impact on trend (slope change) in one effect metric, we modeled the ITS effect estimate halfway through the postintervention period. The difference between the level of the preintervention regression line projected halfway through the postperiod, and the level of the postintervention line at the same time point was used as the ITS effect estimate. This would be comparable with the average effect for the full 12-month period computed in the RCT. We used autoregressive integrated moving average modeling with backward elimination (threshold for retaining coefficients, $P < 0.2$), and took autocorrelation into account by means of an AR1 (autoregressive first order) model.

Last, we also decided to estimate the intervention effect by including data from both study groups and conducting a controlled ITS analysis of the difference between the intervention and control groups in the C-RCT. A monthly time series was constructed for the control group using the same approach as that of the initial ITS analysis of the intervention group only. Subtraction of control group data points from intervention data points, for each month, yielded a time series of data points representing the difference between the two groups. The calculations were done in Stata 12.0 (Stata Corp, College Station, TX) using the arima and xtgee commands.

We assessed the concordance (i.e., consistency) of the estimates qualitatively, taking into consideration the magnitude of the differences in the point estimates and the extent to which the confidence intervals overlapped.

3. Results

Our C-RCT analyses included 139 practices (9,133 prescriptions). The estimated outcome measures in the experimental and control groups for the pre- and postintervention periods are shown graphically in Fig. 2. The estimated intervention effect was an increase of 9.0% (range, 4.9% −13.1%) in the prescribing of recommended antihypertensive medications.

The 70 practices (4,874 prescriptions) in the intervention group were included in the first ITS analysis. The aggregated monthly data points for the intervention group are shown in Fig. 3. The graph supports our assumption that the intervention would have an immediate effect. The change in prescribing during the preintervention period (−0.1%/mo; 95% CI: −0.5, 0.3) and the change in trend from the pre- to the postintervention period (0.3%/mo; 95% CI: −0.3, 0.9) were nonsignificant and therefore not included in the model. Consequently, both the pre- and the postintervention regression lines are flat (Fig. 3). The autocorrelation term was nonsignificant and excluded from the final model. The estimated intervention effect from the ITS analysis was an increase of 11.5% (95% CI: 9.5, 13.5) in the prescribing of recommended antihypertensive medications.

The controlled ITS analysis of the differences between the interventions and control groups is shown graphically in Fig. 4. Again we found an immediate and sustained level change and no change in trend from before to after the intervention. The estimated intervention effect from the controlled ITS analysis was an increase of 14.0% (range, 8.6% −19.4%) in the rate of recommended prescribing.

We also conducted the ITS analyses with and without control group data using generalized estimating equation modeling that included patient-level data and controlled for cluster effects. The results were similar to the findings from the aggregated ITS analyses: 11.9% (95% CI: 8.7, 15.0) for the single-group ITS and 16.1% (95% CI, 9.2, 23.1) for the controlled ITS analysis.

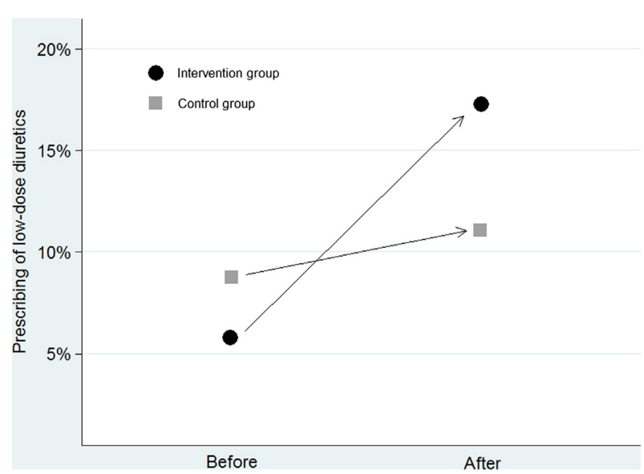

Fig. 2. Prescribing of low-dose diuretics during the pre- and postintervention periods.

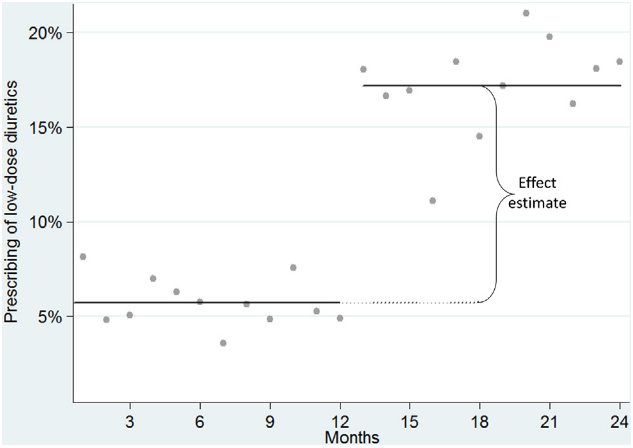

Fig. 3. Monthly prescribing rates of low-dose diuretics with fitted segmented linear regression lines.

4. Discussion

In the case of the Norwegian RaPP trial, a single-group ITS analysis yielded a higher point estimate than the C-RCT analysis (12 percentage points vs. 9 percentage points), but a lower point estimate than the controlled ITS analysis (12 vs. 14). However, the single-group ITS analysis estimate was within the 95% confidence limits of both the other estimates. Thus, in this case, we conclude that the single-group ITS analysis yielded a reliable effect estimate.

This finding is consistent with those of Schneeweiss et al. [7], who found concordant results between a parallel C-RCT and ITS analysis of the effects of a reimbursement restriction of nebulised respiratory therapy in adults. A strength of our comparison is that we reanalyzed before and after data from the same practices in both the C-RCT and the ITS analyses. ITS analyses are widely considered the method of choice for rigorous evaluation of the effects of interventions when RCTs are not feasible [9,10], and our findings are compatible with that view. The feasibility of ITS analysis depends on the availability of routine data of acceptable quality—for example, case records from health care institutions or health insurance claims. An increasing number of routine databases are available that contain clinical and administrative data that could be used for evaluation purposes.

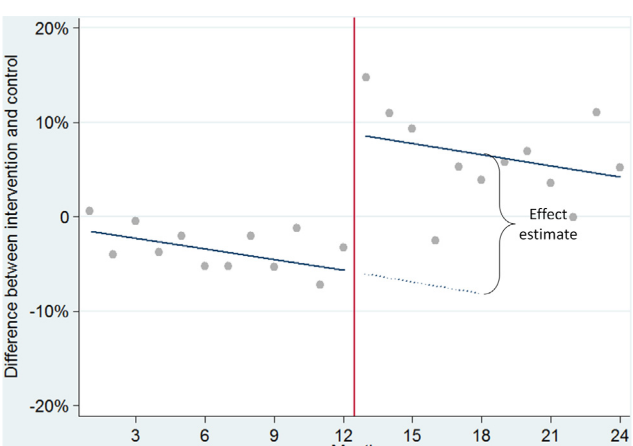

Fig. 4. Difference between the monthly prescribing rates of low-dose diuretics in control and intervention groups, with fitted segmented linear regression lines.

The main weakness of our study is that it is based on a single case. Our findings will require replication in future comparative studies. In some situations, such as the example in the current study, both the RCT and the single-group ITS approach can be used. The main argument favoring an RCT is that the risk of bias is lower because both known and unknown confounders are controlled for through randomization. However, this requires a sufficient sample size, which is difficult to achieve when there are a limited number of clusters that can be randomized. C-RCTs with a small number of clusters may, in fact, have a high risk of bias because of potential confounders. The baseline difference between the groups in the current study (Figs. 2 and 4), resulting from “bad luck randomization,” give reason to question the comparability of the two groups. The single-group ITS likely yields a better estimate if the control group does not serve as a valid comparator. If we do elect to include the control group in the analysis, the question is whether to take into account the difference in trend during the baseline period (Fig. 4), which is the main difference between the C-RCT and controlled ITS analyses. When analyzed separately, neither the control nor the intervention group display a statistically significant trend during the baseline period. However, when the difference between the two groups is used as the dependent variable, the coefficient for trend qualifies for inclusion in the model ($P < 0.2$). The difference between the C-RCT and controlled ITS estimates (9% vs. 14%) is explained by the fact that we took change over time into account in the controlled ITS analysis, but not in the C-RCT analysis. There are no compelling theoretical or practical guidelines to determine which of the three effect estimates is the most valid.

The main risk of bias in ITS analyses is that observed changes may be the result of factors other than the intervention under evaluation. Potential advantages with ITS analyses can be lower costs and greater speed, although this is not always the case [7]. In the case of the RaPP trial, using a single-group ITS analysis would probably have shortened the project period by around 6 months, because half as many practices would have been needed for the study (only intervention practices) compared with a trial that also included control practices.

5. Conclusion

In this study, a single-group ITS analysis yielded an effect estimate similar to a conventional C-RCT analysis or a controlled ITS analysis. A broader range of comparisons from other RCTs would help to determine whether these are generalizable results.

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